Software Requirements Specification (SRS)

Online Bookstore System

Authors: AANYA K (PES2UG24CS011), AARUNI CHOUDHARY (PES2UG24CS013), ADITYA R BADDE (PES2UG25CS802)

Status: Final

Revision history

Version	Date	Author(s)	Change Summary	Approval
1.0	09-09-2026	AANYA K	FR-001 to FR-009: Account + Catalogue + Cart; Section 1 Introduction; Customer UML; NFR-001/RTM entries	—
1.0	09-09-2026	AARUNI CHOUDHARY	FR-010 to FR-017: Checkout + Orders; Section 2 Overall Description; Acceptance-test plan; NFR-002/003; Security Objective 1 + SR-001/002; RTM entries	—
1.0	09-09-2026	ADITYA R BADDE	FR-018 to FR-026: Orders + Payment + Administration + Inventory; Section 3 External Interfaces; Administrator UML; NFR-004 to NFR-006; Security Objective 2 + SR-003 to SR-006; RTM entries	—

Approvals

Role	Name	Signature / Email	Date
Course Coordinator / Guide	Prof Pavithra P		

Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features
5. Non-functional requirements
    5.1 Security
    5.1.1 Security Objectives
    5.1.2 Security Requirements
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Online Bookstore System. It defines the functional and non-functional requirements, external interfaces, and verification criteria for a system that allows customers to browse and purchase books and allows administrators to manage the bookstore.

1.2 Scope

The system covers customer registration and login, book browsing and searching, shopping cart management, checkout, online payment, order confirmation, order history and order status. It also covers administration of books, categories, stock and orders, together with basic sales and inventory reports. The internal operation of external payment and delivery services is outside the scope of this system.

1.3 Audience

The intended readers are the student development team, project guide, evaluators, testers, system administrator and other stakeholders who need to understand the expected behaviour of the bookstore.

1.4 Definitions

- SRS - Software Requirements Specification
- FR - Functional Requirement
- NFR - Non-functional Requirement
- UI - User Interface
- API - Application Programming Interface
- ISBN - International Standard Book Number
- RBAC - Role-Based Access Control
- RTM - Requirements Traceability Matrix
- TLS - Transport Layer Security

2. Overall description

2.1 Product perspective

The Online Bookstore is a web-based system used by customers and bookstore administrators. Customers use the system to find books, add them to a cart and place orders. Administrators use it to maintain book information, stock and orders. The system communicates with an external payment service when an online payment is made.

2.2 Major product functions

- Customer registration and authentication
- Book catalogue browsing
- Book search, filtering and sorting
- Book detail display
- Shopping cart management
- Checkout and delivery information
- Order total calculation
- Online payment and payment result handling
- Order confirmation and order history
- Order status viewing and cancellation
- Book and category management
- Inventory and low-stock management
- Order administration
- Basic sales and inventory reports

2.3 User roles and characteristics

- Customer: A normal user who searches for books, purchases books and views their own order information. Basic web browsing knowledge is expected.
- Administrator: A user who maintains books, categories, stock and orders and can view bookstore reports. Administrator functions require authorization.
- Payment Service: An external service that receives payment requests and returns a payment result. It is not controlled by the bookstore system.

2.4 Operating environment

The system shall be accessible through commonly used web browsers on desktop, laptop, tablet and mobile devices. A stable internet connection is required for normal use and online payment. The system requires persistent storage for customer, book, inventory, cart, order and payment information.

2.5 Constraints

- The system depends on an external payment service for online payment authorization.
- Only authorized administrators shall access administrative functions.
- Stock information must remain consistent with confirmed orders.
- The system shall not require proprietary browser plugins.
- The project shall remain within the time and resources available to the student team.

2.6 Assumptions and dependencies

- Book information and stock quantities are maintained by an administrator.
- Customers provide valid account and delivery information.
- The payment service is available when a payment is attempted.
- Physical packing and delivery are handled outside the bookstore system.

3. External interfaces

3.1 User interfaces

The customer interface shall provide a clear menu for browsing books, searching, viewing book details, managing the cart, checking out and viewing orders. The administrator interface shall provide access to books, categories, inventory, orders and reports. Forms shall show clear messages when input is missing or invalid.

3.2 Hardware interfaces

- Standard keyboard and mouse input on desktop or laptop systems.
- Touch input on supported tablet and mobile devices.
- A display capable of showing the web interface.
- A standard printer may be used by the user when an order summary or report is printed.

3.3 Software interfaces

- Application interface between the user interface and bookstore services.
- Data storage for customer, book, category, inventory, cart, order and payment records.
- External payment service for online payment authorization and transaction status.
- Optional notification service for order confirmation and status messages.

3.4 Communications

- Account, order and payment-related communication shall use secure network connections.
- The system shall verify the result returned by the payment service before an order is marked as paid.
- A communication failure shall not cause an order to be incorrectly marked as paid.

4. System features

Each requirement below includes acceptance criteria and a reference test case. The requirements are written as specific system behaviour so that they can be checked during validation.

4.1 Authentication and customer account

Provide basic account access for customers.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-001	The system shall allow a customer to create an account using a unique email address and password.	Functional	High	Customer	AC-OBS-F-001: Valid details create an account; an existing email is rejected. Test: TC-Auth-01	Validation required.
OBS-F-002	The system shall allow a registered customer to log in using valid credentials.	Functional	High	Customer	AC-OBS-F-002: Correct credentials allow access and incorrect credentials are rejected. Test: TC-Auth-02	Authentication required.
OBS-F-003	The system shall allow an authenticated customer to log out.	Functional	High	Customer	AC-OBS-F-003: Logout ends the session and protected account pages require login again. Test: TC-Auth-03	—

4.2 Book catalogue and search

Allow customers to find books and view current catalogue information.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-004	The system shall display active books with title, author, price, category and availability.	Functional	High	Customer	AC-OBS-F-004: The catalogue displays the required information for active books. Test: TC-Cat-01	Book data required.
OBS-F-005	The system shall allow customers to search books by title, author or ISBN.	Functional	High	Customer	AC-OBS-F-005: Matching books are returned and a no-results message is shown when appropriate. Test: TC-Cat-02	Search data required.
OBS-F-006	The system shall allow customers to filter books by category and price range.	Functional	Medium	Customer	AC-OBS-F-006: Results contain only books matching the selected filters. Test: TC-Cat-03	—
OBS-F-007	The system shall allow customers to sort displayed books by title or price.	Functional	Medium	Customer	AC-OBS-F-007: Results are ordered correctly for the selected option. Test: TC-Cat-04	—
OBS-F-008	The system shall display detailed information for a selected book, including description, author, ISBN, price and availability.	Functional	High	Customer	AC-OBS-F-008: The selected book opens with current stored details. Test: TC-Book-01	—

4.3 Shopping cart and checkout

Allow customers to prepare an order before payment.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-009	The system shall allow a customer to add an available book and quantity to the shopping cart.	Functional	High	Customer	AC-OBS-F-009: The selected book and quantity appear in the cart with the correct subtotal. Test: TC-Cart-01	Stock validation required.
OBS-F-010	The system shall allow a customer to change the quantity of an item or remove an item from the cart.	Functional	High	Customer	AC-OBS-F-010: The cart reflects the change and recalculates the total. Test: TC-Cart-02	—
OBS-F-011	The system shall prevent a customer from adding a quantity greater than available stock.	Functional	High	Customer	AC-OBS-F-011: A quantity above stock is rejected with a clear message. Test: TC-Cart-03	Inventory data required.
OBS-F-012	The system shall collect the delivery address and contact details required to place an order.	Functional	High	Customer	AC-OBS-F-012: Checkout does not continue when mandatory information is missing or invalid. Test: TC-Order-01	Input validation required.
OBS-F-013	The system shall calculate the order total from item prices, quantities, delivery charges and applicable discounts.	Functional	High	Customer	AC-OBS-F-013: The displayed total matches the applicable order values. Test: TC-Order-02	Pricing rules required.

4.4 Orders and payment

Create, pay for and track customer orders.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-014	The system shall send the payment request for an order to the external payment service.	Functional	High	Customer / Payment Service	AC-OBS-F-014: A payment request is submitted with the order amount and a result is received. Test: TC-Pay-01	External payment dependency.
OBS-F-015	The system shall record payment as successful only after a valid successful response is received from the payment service.	Functional	High	Customer / Payment Service	AC-OBS-F-015: Successful response marks payment successful; failed response does not. Test: TC-Pay-02	Payment result must be checked.
OBS-F-016	The system shall create a unique order number after successful order confirmation.	Functional	High	Customer	AC-OBS-F-016: Each confirmed order receives a unique order number and confirmation. Test: TC-Order-03	—
OBS-F-017	The system shall reduce available stock after an order is successfully confirmed.	Functional	High	Administrator	AC-OBS-F-017: Stock is reduced by confirmed quantities; failed payment does not reduce stock. Test: TC-Inv-01	Order and stock records must agree.
OBS-F-018	The system shall allow a customer to view their order history and current order status.	Functional	High	Customer	AC-OBS-F-018: A customer sees only their own orders and their stored statuses. Test: TC-Order-04	Authentication required.
OBS-F-019	The system shall allow a customer to cancel an order before the order reaches a non-cancellable status.	Functional	Medium	Customer	AC-OBS-F-019: Eligible order is cancelled; non-eligible order is rejected with a message. Test: TC-Order-05	Cancellation rule required.

4.5 Administration and inventory

Allow authorized administrators to maintain bookstore information and stock.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-020	The system shall allow an authorized administrator to add a new book with required catalogue information.	Functional	High	Administrator	AC-OBS-F-020: Valid book information is saved and the book appears in the catalogue when active. Test: TC-Admin-01	Administrator access required.
OBS-F-021	The system shall allow an authorized administrator to edit book details and deactivate a book.	Functional	High	Administrator	AC-OBS-F-021: Changes are saved and a deactivated book is not available for normal purchase. Test: TC-Admin-02	Administrator access required.
OBS-F-022	The system shall allow an authorized administrator to create and update book categories.	Functional	Medium	Administrator	AC-OBS-F-022: Valid category changes are saved and reflected in catalogue filtering. Test: TC-Admin-03	—
OBS-F-023	The system shall allow an authorized administrator to update book stock quantity and reorder level.	Functional	High	Administrator	AC-OBS-F-023: Updated values are saved and reflected in availability. Test: TC-Inv-02	Inventory data required.
OBS-F-024	The system shall identify books whose stock quantity is at or below the reorder level.	Functional	Medium	Administrator	AC-OBS-F-024: All books at or below the threshold are identified. Test: TC-Inv-03	Reorder level required.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
OBS-F-025	The system shall allow an authorized administrator to update the status of an order.	Functional	High	Administrator	AC-OBS-F-025: Authorized status changes are saved and visible in the customer's order view. Test: TC-Admin-04	Status values required.
OBS-F-026	The system shall allow an authorized administrator to view basic sales and inventory reports for a selected period.	Functional	Medium	Administrator	AC-OBS-F-026: Report totals match the relevant stored records. Test: TC-Rep-01	Reporting data required.

5. Non-functional requirements (detailed)

The NFRs are measurable where practical and are linked to a test or review method.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
OBS-NF-001	Catalogue, search and book-detail pages shall load within 3 seconds for at least 90% of requests under normal project test load.	Performance	High	90th percentile response time ≤ 3 seconds. Test: TC-Perf-01
OBS-NF-002	The system shall provide at least 99.5% availability per month, excluding planned maintenance.	Reliability	High	Availability report shows ≥ 99.5%. Test: TC-Reliability-01
OBS-NF-003	A new customer shall be able to search for a book and add it to the cart without formal training.	Usability	Medium	At least 4 of 5 first-time users complete the task without assistance. Test: TC-UX-01
OBS-NF-004	The system shall support at least 10,000 book records and 20 concurrent customer sessions without failing the normal performance target.	Scalability	Medium	Load test with 10,000 records and 20 sessions passes. Test: TC-Scale-01
OBS-NF-005	The system shall be usable on desktop, tablet and mobile screen sizes for the main customer functions.	Portability / Usability	Medium	Browse, search, cart and checkout work at selected viewport sizes. Test: TC-UX-02
OBS-NF-006	The system shall retain records of administrator changes to books, inventory and order status.	Auditability	Medium	Sample audit records contain user, action, affected record and time. Test: TC-Audit-01

5.1 Security

5.1.1 Security Objectives

The security objectives are based on the main assets and risks of the bookstore system.

Objective ID	Security objective	Risk addressed	Measure
OBS-SO-001	Protect customer account and order information from unauthorized access.	Unauthorized access to customer information	Protected functions require authentication and authorization; security tests pass.
OBS-SO-002	Protect payment and order information from unauthorized modification.	False payment confirmation or unauthorized order changes	Only valid payment results change payment status; unauthorized changes are denied.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OBS-SR-001	The system shall require authentication before a user can access protected account or administrator functions.	Security	High	AC-OBS-SR-001: Unauthenticated access is rejected. Test: TC-Sec-01
OBS-SR-002	The system shall store user passwords using a salted one-way hashing method and shall not store passwords as plaintext.	Security	High	AC-OBS-SR-002: Stored password values are hashes rather than plaintext. Test: TC-Sec-02

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
OBS-SR-003	The system shall enforce role-based access control so that customers cannot perform administrator functions.	Security	High	AC-OBS-SR-003: Customer access to an administrator function is denied. Test: TC-Sec-03
OBS-SR-004	The system shall use secure communication for client-server traffic containing account, order or payment-related information.	Security	High	AC-OBS-SR-004: Protected traffic uses a secure connection. Test: TC-Sec-04
OBS-SR-005	The system shall validate and sanitize input used in account, search, address and administrator forms.	Security	High	AC-OBS-SR-005: Tested invalid and malicious inputs do not cause unauthorized database or script execution. Test: TC-Sec-05
OBS-SR-006	The system shall verify the payment-service response before marking an order as paid.	Security	High	AC-OBS-SR-006: Invalid or altered payment results do not mark an order as paid. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

6.1 Quality attributes

Attribute	Expected quality
Performance	Common customer pages respond within the response-time target under normal load.
Reliability	Confirmed order information remains available after normal service recovery.
Usability	A new customer can complete basic browsing and cart tasks without formal training.
Scalability	The system supports the specified number of book records and concurrent sessions.
Security	Protected information and administrator operations are available only to authorized users.
Maintainability	Requirements, design references and test cases remain traceable when changes are made.

6.2 Acceptance tests

Acceptance testing will use the test case references attached to the requirements. The tests should cover normal input, invalid input and important boundary cases. High-priority requirements should be verified before the system is considered ready for acceptance.

Test suite	Coverage	Test cases	Method
Authentication	Registration, login, logout and access control	TC-Auth-01 to TC-Auth-03	Functional and security testing
Catalogue	Browse, search, filter, sort and book details	TC-Cat-01 to TC-Book-01	Functional testing
Cart and checkout	Cart changes, stock check, address and total	TC-Cart-01 to TC-Order-02	Functional and boundary testing
Payment and orders	Payment result, confirmation, stock, status and cancellation	TC-Pay-01 to TC-Order-05	Integration and functional testing
Administration	Books, categories, inventory, orders and reports	TC-Admin-01 to TC-Rep-01	Functional and authorization testing
Quality	Performance, reliability, usability, scalability and audit	TC-Perf-01, TC-Reliability-01, TC-UX-01/02, TC-Scale-01, TC-Audit-01	Measurement and review
Security	Authentication, password protection, authorization, input validation, session and payment integrity	TC-Sec-01 to TC-Sec-06	Security testing and review

6.3 Acceptance exit criteria

The project is ready for acceptance when all high-priority functional requirements have been implemented and verified, there are no unresolved critical NFR or security failures, and the RTM provides test coverage for all requirements.

7. UML Use-Case Diagram

7.1 Customer use-case diagram

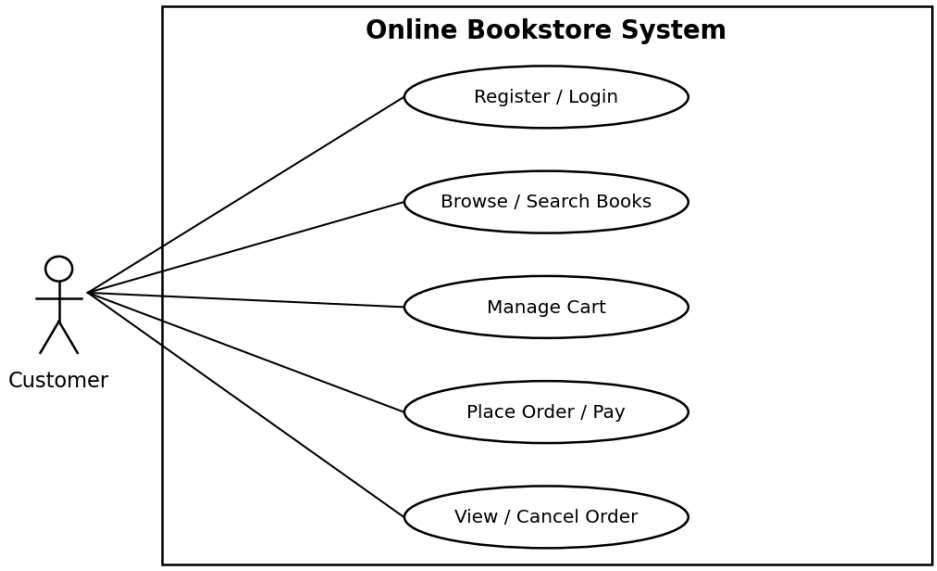


Figure 1: Customer use-case diagram

7.2 Administrator use-case diagram

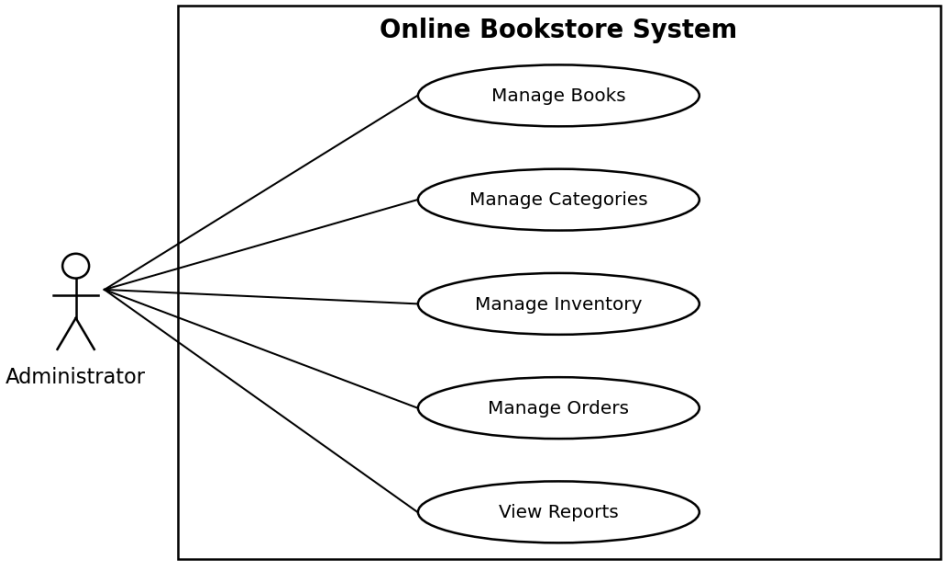


Figure 2: Administrator use-case diagram

8. Requirements Traceability Matrix (RTM)

The RTM links all functional, non-functional and security requirements to the relevant SRS section, design area, module and planned test case. Status is N (not started), P (partially implemented) or A (accepted). At the requirements stage the planned status is N.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status	Comments
OBS-F-001	Allow customer to create an account using a unique email address and password	4.1 / DS-F001	Authentication	TC-Auth-01	N	
OBS-F-002	allow a registered customer to log in using valid credentials	4.1 / DS-F002	Authentication	TC-Auth-02	N	
OBS-F-003	allow an authenticated customer to log out	4.1 / DS-F003	Authentication	TC-Auth-03	N	
OBS-F-004	Display active books with title, author, price, category and availability	4.2 / DS-F004	Catalogue	TC-Cat-01	N	
OBS-F-005	allow customers to search books by title, author or ISBN	4.2 / DS-F005	Catalogue	TC-Cat-02	N	
OBS-F-006	allow customers to filter books by category and price range	4.2 / DS-F006	Catalogue	TC-Cat-03	N	
OBS-F-007	allow customers to sort displayed books by title or price	4.2 / DS-F007	Catalogue	TC-Cat-04	N	
OBS-F-008	Display detailed information for a selected book, including description, author, ISBN, price and availability	4.2 / DS-F008	Catalogue	TC-Book-01	N	
OBS-F-009	Allow customer to add an available book and quantity to the shopping cart	4.3 / DS-F009	Cart / Checkout	TC-Cart-01	N	
OBS-F-010	Allow customer to change item quantity or remove an item from the cart	4.3 / DS-F010	Cart / Checkout	TC-Cart-02	N	
OBS-F-011	prevent a customer from adding a quantity greater than available stock	4.3 / DS-F011	Cart / Checkout	TC-Cart-03	N	
OBS-F-012	Collect delivery address and contact details required to place an order .	4.3 / DS-F012	Cart / Checkout	TC-Order-01	N	
OBS-F-013	Calculate order total from item prices, quantities, delivery charges and applicable discounts	4.3 / DS-F013	Cart / Checkout	TC-Order-02	N	
OBS-F-014	send the payment request for an order to the external payment service	4.4 / DS-F014	Payment / Orders / Inventory	TC-Pay-01	N	
OBS-F-015	Record payment as successful only after receiving a valid successful response from the payment service .	4.4 / DS-F015	Payment / Orders / Inventory	TC-Pay-02	N	
OBS-F-016	create a unique order number after successful order confirmation	4.4 / DS-F016	Payment / Orders / Inventory	TC-Order-03	N	
OBS-F-017	reduce available stock after an order is successfully confirmed	4.4 / DS-F017	Payment / Orders / Inventory	TC-Inv-01	N	
OBS-F-018	allow a customer to view their order history and current order status	4.4 / DS-F018	Payment / Orders / Inventory	TC-Order-04	N	
OBS-F-019	Allow customer to cancel an order before it reaches a non-cancellable status .	4.4 / DS-F019	Payment / Orders / Inventory	TC-Order-05	N	
OBS-F-020	Allow authorized administrator to add a new book with required catalogue information .	4.5 / DS-F020	Administration / Inventory / Reporting	TC-Admin-01	N	
OBS-F-021	Allow authorized administrator to edit book details and deactivate a book	4.5 / DS-F021	Administration / Inventory / Reporting	TC-Admin-02	N	
OBS-F-022	allow an authorized administrator to create and update book categories	4.5 / DS-F022	Administration / Inventory / Reporting	TC-Admin-03	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status	Comments
OBS-F-023	Allow authorized administrator to update book stock quantity and reorder level	4.5 / DS-F023	Administration / Inventory / Reporting	TC-Inv-02	N	
OBS-F-024	identify books whose stock quantity is at or below the reorder level	4.5 / DS-F024	Administration / Inventory / Reporting	TC-Inv-03	N	
OBS-F-025	allow an authorized administrator to update the status of an order	4.5 / DS-F025	Administration / Inventory / Reporting	TC-Admin -04	N	
OBS-F-026	Allow authorized administrator to view basic sales and inventory reports for a selected period	4.5 / DS-F026	Administration / Inventory / Reporting	TC-Rep-01	N	
OBS-NF-00 1	Catalogue, search and book-detail pages shall load within 3 seconds for at least 90% of requests under normal project test load	5 / DS-NF00 1	Web UI / Services	TC-Perf-01	N	
OBS-NF-00 2	Provide at least 99.5% availability per month, excluding planned maintenance	5 / DS-NF00 2	System / Services	TC-Reliabi lity-01	N	
OBS-NF-00 3	Allow a new customer to search for a book and add it to the cart without formal training	5 / DS-NF00 3	Customer UI	TC-UX-01	N	
OBS-NF-00 4	Support at least 10,000 book records and 20 concurrent customer sessions without failing the normal performance target	5 / DS-NF00 4	Web UI / Data	TC-Scale-0 1	N	
OBS-NF-00 5	Be usable on desktop, tablet and mobile screen sizes for the main customer functions	5 / DS-NF00 5	Customer UI	TC-UX-02	N	
OBS-NF-00 6	Retain records of administrator changes to books, inventory and order status	5 / DS-NF00 6	Audit / Administration	TC-Audit-01	N	
OBS-SR-00 1	Require authentication before a user can access protected account or administrator functions	5.1.2 / DS-SR00 1	Authentication / Authorization	TC-Sec-01	N	
OBS-SR-00 2	Store user passwords using a salted one-way hashing method and not as plaintext	5.1.2 / DS-SR00 2	Authentication	TC-Sec-02	N	
OBS-SR-00 3	Enforce role-based access control so customers cannot perform administrator functions	5.1.2 / DS-SR00 3	Authorization	TC-Sec-03	N	
OBS-SR-00 4	Use secure communication for client-server traffic containing account, order or payment-related information	5.1.2 / DS-SR00 4	Communication	TC-Sec-04	N	
OBS-SR-00 5	Validate and sanitize input used in account, search, address and administrator forms	5.1.2 / DS-SR00 5	Input Validation	TC-Sec-05	N	
OBS-SR-00 6	Verify the payment-service response before marking an order as paid	5.1.2 / DS-SR00 6	Payment	TC-Sec-06	N	